

Example 32.1. The maximum flux density in the core of a 250/3000-volts, 50-Hz single-phase transformer is 1.2 Wb/m^2 . If the e.m.f. per turn is 8 volt, determine

- (i) primary and secondary turns (ii) area of the core.

(Electrical Engg.-I, Nagpur Univ. 1991)

Solution. (i)

$$E_1 = N_1 \times \text{e.m.f. induced/turn}$$

$$N_1 = 250/8 = 32; N_2 = 3000/8 = 375$$

(ii) We may use

$$E_2 = -4.44 f N_2 B_m A$$

$\therefore$

$$3000 = 4.44 \times 50 \times 375 \times 1.2 \times A; A = 0.03 \text{ m}^2.$$

Example 32.2. The core of a 100-kVA, 11000/550 V, 50-Hz, 1-ph, core type transformer has a cross-section of $20 \text{ cm} \times 20 \text{ cm}$. Find (i) the number of H.V. and L.V. turns per phase and (ii) the e.m.f. per turn if the maximum core density is not to exceed 1.3 Tesla. Assume a stacking factor of 0.9.

What will happen if its primary voltage is increased by 10% on no-load ?

(Elect. Machines, A.M.I.E. Sec. B, 1991)

Solution. (i)

$$B_m = 1.3 \text{ T}, A = (0.2 \times 0.2) \times 0.9 = 0.036 \text{ m}^2$$

$\therefore$

$$11,000 = 4.44 \times 50 \times N_1 \times 1.3 \times 0.036, N_1 = 1060$$

$$550 = 4.44 \times 50 \times N_2 \times 1.3 \times 0.036; N_2 = 53$$

or,

$$N_2 = K N_1 = (550/11,000) \times 1060 = 53$$

(ii)

$$\text{e.m.f./turn} = 11,000/1060 = 10.4 \text{ V or } 550/53 = 10.4 \text{ V}$$

Keeping supply frequency constant, if primary voltage is increased by 10%, magnetising current will increase by much more than 10%. However, due to saturation, flux density will increase only marginally and so will the eddy current and hysteresis losses.

Example 32.3. A single-phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm^2 . If the primary winding be connected to a 50-Hz supply at 520 V, calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding.

(Elect. Engg.-I, Pune Univ. 1989)

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Solution.

$$K = N_2/N_1 = 1000/400 = 2.5$$

(i)

$$E_2/E_1 = K \therefore E_2 = K E_1 = 2.5 \times 520 = 1300 \text{ V}$$

(ii)

$$E_1 = 4.44 f N_1 B_m A$$

or

$$520 = 4.44 \times 50 \times 400 \times B_m \times (60 \times 10^{-4}) \therefore B_m = 0.976 \text{ Wb/m}^2$$

Example 32.4. A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current.

(Elect. & Electronic Engg., Madras Univ. 1985)

Solution.

$$K = N_2/N_1 = 50/500 = 1/10$$

Now, full-load

$$I_1 = 25,000/3000 = 8.33 \text{ A. F.L. } I_2 = I_1/K = 10 \times 8.33 = 83.3 \text{ A}$$

e.m.f. per turn on primary side = $3000/500 = 6 \text{ V}$

$\therefore$

$$\text{secondary e.m.f.} = 6 \times 50 = 300 \text{ V (or } E_2 = K E_1 = 3000 \times 1/10 = 300 \text{ V)}$$

Also,

$$E_1 = 4.44 f N_1 \Phi_m; 3000 = 4.44 \times 50 \times 500 \times \Phi_m \therefore \Phi_m = 27 \text{ mWb}$$

Example 32.15. A 30 kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of 0.1Ω and a leakage reactance of 0.22Ω . The low voltage winding resistance is 0.035Ω and the leakage reactance is 0.012Ω . Find the equivalent winding resistance, reactance and impedance referred to the (i) high voltage side and (ii) the low-voltage side.

(Electrical Machines-I, Bangalore Univ. 1987)

Solution.

$$K = 120/2400 = 1/20; R_1 = 0.1 \Omega, X_1 = 0.22 \Omega$$

$$R_2 = 0.035 \Omega \quad \text{and} \quad X_2 = 0.012 \Omega$$

(i) Here, high-voltage side is, obviously, the primary side. Hence, values as referred to primary side are

$$R_{01} = R_1 + R_2' = R_1 + R_2/K^2 = 0.1 + 0.035/(1/20)^2 = \mathbf{14.1 \Omega}$$

$$X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 0.22 + 0.012/(1/20)^2 = \mathbf{5.02 \Omega}$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{14.1^2 + 5.02^2} = \mathbf{15 \Omega}$$

(ii)

$$R_{02} = R_2 + R_1' = R_2 + K^2 R_1 = 0.035 + (1/20)^2 \times 0.1 = \mathbf{0.03525 \Omega}$$

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.012 + (1/20)^2 \times 0.22 = \mathbf{0.01255 \Omega}$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.03525^2 + 0.01255^2} = \mathbf{0.0374 \Omega}$$

$$(\text{or } Z_{02} = K^2 Z_{01} = (1/20)^2 \times 15 = 0.0375 \Omega)$$

Example 32.16. A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The values of reactances are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. Calculate for the transformer (i) equivalent resistance as referred to primary (ii) equivalent resistance as referred to secondary (iii) equivalent reactance as referred to both primary and secondary (iv) equivalent impedance as referred to both primary and secondary (v) total Cu loss, first using individual resistances of the two windings and secondly, using equivalent resistances as referred to each side.

(Elect. Engg.-I, Nagpur Univ. 1993)

Solution. Full-load

$$I_1 = 50,000/4,400 = 11.36 \text{ A (assuming 100\% efficiency)}$$

Full-load

$$I_2 = 50,000/220 = 227 \text{ A}; K = 220/4,400 = 1/20$$

(i)

$$R_{01} = R_1 + \frac{R_2}{K^2} = 3.45 + \frac{0.009}{(1/20)^2} = 3.45 + 3.6 = \mathbf{7.05 \Omega}$$

(ii)

$$R_{02} = R_2 + K^2 R_1 = 0.009 + (1/20)^2 \times 3.45 = 0.009 + 0.0086 = \mathbf{0.0176 \Omega}$$

Also,

$$R_{02} = K^2 R_{01} = (1/20)^2 \times 7.05 = 0.0176 \Omega \text{ (check)}$$

(iii)

$$X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 5.2 + 0.015/(1/20)^2 = \mathbf{11.2 \Omega}$$

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.015 + 5.2/20^2 = \mathbf{0.028 \Omega}$$

Also X_{02}

$$= K^2 X_{01} = 11.2/400 = 0.028 \Omega \text{ (check)}$$

(iv)

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{7.05^2 + 11.2^2} = \mathbf{13.23 \Omega}$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.0176^2 + 0.028^2} = \mathbf{0.03311 \Omega}$$

Also Z_{02}

$$= K^2 Z_{01} = 13.23/400 = 0.0331 \Omega \text{ (check)}$$

(v)

$$\text{Cu loss} = I_1^2 R_1 + I_2^2 R_2 = 11.36^2 \times 3.45 + 227^2 \times 0.009 = \mathbf{910 \text{ W}}$$

Also Cu loss

$$= I_1^2 R_{01} = 11.36^2 \times 7.05 = \mathbf{910 \text{ W}}$$

$$= I_2^2 R_{02} = 227^2 \times 0.0176 = \mathbf{910 \text{ W}}$$

Example 32.21. A 230/460-V transformer has a primary resistance of $0.2\ \Omega$ and reactance of $0.5\ \Omega$ and the corresponding values for the secondary are $0.75\ \Omega$ and $1.8\ \Omega$ respectively. Find the secondary terminal voltage when supplying 10 A at 0.8 p.f. lagging .

(Electric Machines-II, Bangalore Univ. 1991)

Solution.

$$K = 460/230 = 2; R_{02} = R_2 + K^2 R_1 = 0.75 + 2^2 \times 0.2 = 1.55\ \Omega$$

$$X_{02} = X_2 + K^2 X_1 = 1.8 + 2^2 \times 0.5 = 3.8\ \Omega$$

$$\text{Voltage drop} = I_2 (R_{02} \cos \phi + X_{02} \sin \phi) = 10 (1.55 \times 0.8 + 3.8 \times 0.6) = 35.2\text{ V}$$

$$\therefore \text{Secondary terminal voltage} = 460 - 35.2 = \mathbf{424.8\text{ V}}$$

Example 32.22. Calculate the regulation of a transformer in which the percentage resistance drops is 1.0% and percentage reactance drop is 5.0% when the power factor is (a) 0.8 lagging (b) unity and (c) 0.8 leading. (Electrical Engineering, Banaras Hindu Univ. 1988)

Solution. We will use the approximate expression of Art 30.16.

$$(a) \text{ p.f.} = \cos \phi = 0.8 \text{ lag} \quad \mu = v_r \cos \phi + v_x \sin \phi = 1 \times 0.8 + 5 \times 0.6 = \mathbf{3.8\%}$$

$$(b) \text{ p.f.} = \cos \phi = 1 \quad \mu = 1 \times 1 + 5 \times 0 = \mathbf{1\%}$$

$$(c) \text{ p.f.} = \cos \phi = 0.8 \text{ lead} \quad \mu = 1 \times 0.8 - 5 \times 0.6 = \mathbf{-2.2\%}$$

Example 32.27. In no-load test of single-phase transformer, the following test data were obtained :

Primary voltage : 220 V ; Secondary voltage : 110 V ;

Primary current : 0.5 A ; Power input : 30 W .

Find the following :

(i) The turns ratio (ii) the magnetising component of no-load current (iii) its working (or loss) component (iv) the iron loss.

Resistance of the primary winding = 0.6 ohm .

Draw the no-load phasor diagram to scale.

(Elect. Machine A.M.I.E. 1990)

Solution. (i) Turn ratio, $N_1/N_2 = 220/110 = 2$

$$(ii) W = V_1 I_0 \cos \phi_0; \cos \phi_0 = 30/220 \times 0.5 = 0.273; \sin \phi_0 = 0.962$$

$$I_\mu = I_0 \sin \phi_0 = 0.5 \times 0.962 = 0.48\text{ A}$$

$$(iii) I_w = I_0 \cos \phi_0 = 0.5 \times 0.273 = 0.1365\text{ A}$$

$$(iv) \text{ Primary Cu loss} = I_0^2 R_1 = 0.5^2 \times 0.6 = 0.15\text{ W}$$

$$\therefore \text{Iron loss} = 30 - 0.15 = 29.85\text{ W}$$

Example 32.32. When a transformer is connected to a 1000-V , 50-Hz supply the core loss is 1000 W , of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to $2,000\text{ V}$ and the frequency to 100 Hz , find the new core losses.

$$\text{Solution. Hysteresis loss } W_h \propto B_{\max}^{1.6} f = P B_{\max}^{1.6} f$$

$$\text{Eddy current loss } W_e \propto B_{\max}^2 f^2 = Q B_{\max}^2 f^2$$

$$\text{From the relation } E = 4.44 f N B_{\max} \text{ A volt, we get } B_{\max} \propto E/f$$

Putting this value of $B_{\max}$ in the above equations, we have

$$W_h = P \left(\frac{E}{f} \right)^{1.6} f = P E^{1.6} f^{-0.6} \text{ and } W_e = Q \left(\frac{E}{f} \right)^2 f^2 = Q E^2$$

In the first case,

$$E = 1000\text{ V}, f = 50\text{ Hz}, W_h = 650\text{ W}, W_e = 350\text{ W}$$

$$\therefore 650 = P \times 1000^{1.6} \times 50^{-0.6} \quad \therefore P = 650 \times 1000^{-1.6} \times 50^{0.6}$$

Similarly,

$$350 = Q \times 1000^2 \quad \therefore Q = 350 \times 1000^{-2}$$

Hence, constants P and Q are known.

Using them in the second case, we get

$$W_h = (650 \times 1000^{-1.6} \times 50^{0.6}) \times 2000^{1.6} \times 100^{-0.6} = 650 \times 2 = 1,300 \text{ W}$$

$$W_e = (350 \times 1000^{-2}) \times 2,000^2 = 350 \times 4 = 1,400 \text{ W}$$

$\therefore$ Core loss under new condition is $= 1,300 + 1,400 = 2700 \text{ W}$

Alternative Solution

Here, both voltage and frequency are doubled, leaving the flux density unchanged.

With 1000 V at 50 Hz

$$W_h = Af \text{ or } 650 = 50 A ; A = 13$$

$$W_e = Bf^2 \text{ or } 350 = B \times 50^2 ; B = 7/50$$

With 2000 V at 100 Hz

$$W_h = Af = 13 \times 100 = 1300 \text{ W and}$$

$$W_e = Bf^2 = (7/50) \times 100^2 = 1400 \text{ W}$$

$\therefore$ New core loss $= 1300 + 1400 = 2700 \text{ W}$

Example 32.35. The primary and secondary windings of a 30 kVA 76000/230, V, 1-phase transformer have resistance of 10 ohm and 0.016 ohm respectively. The reactance of the transformer referred to the primary is 34 ohm. Calculate the primary voltage required to circulate full-load current when the secondary is short-circuited. What is the power factor on short circuit ?

(Elect. Machines AMIE Sec. B 1991)

Solution.

$$K = 230/6000 = 23/600, X_{01} = 34 \Omega,$$

$$R_{01} = R_1 + R_2/K^2 = 10 + 0.016 (600/23)^2 = 20.9 \Omega$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{20.9^2 + 34^2} = 40 \Omega$$

$$\text{F.L., } I_1 = 30,000/6000 = 5 \text{ A ; } V_{SC} = I_1 Z_{01} = 5 \times 40 = 200 \text{ V}$$

$$\text{Short circuit p.f.} = R_{01}/Z_{01} = 20.9/40 = 0.52$$